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1. a) In the following matrix-vector multiplication over the real numbers, find the values of a and b .

$$\begin{pmatrix} 1 & 0 & 4 & 0 \\ 2 & 1 & 5 & a \\ 3 & 2 & 6 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 12 \\ b \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 4 & 0 \\ 2 & 1 & 5 & a \\ 3 & 2 & 6 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 + 4 \cdot 0 + 0 \cdot 5 \\ 2 \cdot 1 + 1 \cdot 0 + 5 \cdot 0 + a \cdot 5 \\ 3 \cdot 1 + 2 \cdot 0 + 6 \cdot 0 + 6 \cdot 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 5a+2 \\ 33 \end{pmatrix}$$

Since $\begin{pmatrix} 1 \\ 5a+2 \\ 33 \end{pmatrix} = \begin{pmatrix} 1 \\ 12 \\ b \end{pmatrix}$, we have

$$\boxed{\begin{matrix} a=2 \\ b=33 \end{matrix}}$$

- b) Let B be the following real matrix:

$$B = \begin{pmatrix} 0 & 1 & 2 & 3 & 4 & 0 \\ 0 & 4 & 3 & 2 & 1 & 8 \end{pmatrix}$$

Complete the following sentence by filling in the four blanks:

The matrix B is a 2 $\times$ 6 matrix and it represents a linear transformation from $\mathbb{R}^{\underline{6}}$ to $\mathbb{R}^{\underline{2}}$.

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2. a) For each of the following matrix operations, calculate the result if the operation is defined. Otherwise, state that the operation is undefined.

$$A = \begin{pmatrix} 2 & 0 & 3 & 0 \\ 0 & 1 & -1 & 4 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 \\ 4 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 3 & -2 \\ 0 & 1 \end{pmatrix}$$

$C - B$, CB , AD , DA

$$C - B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} 0-2 & 0-0 \\ 0-4 & 1-1 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ -4 & 0 \end{pmatrix}$$

$$CB = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} 0 \cdot 2 + 0 \cdot 4 & 0 \cdot 0 + 0 \cdot 1 \\ 0 \cdot 2 + 1 \cdot 4 & 0 \cdot 0 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 4 & 1 \end{pmatrix}$$

AD is not defined.

$$DA = \begin{pmatrix} 0 & 0 \\ 3 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 3 & 0 \\ 0 & 1 & -1 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 6 & -2 & 11 & -8 \\ 0 & 1 & -1 & 4 \end{pmatrix}$$

b) Choose an integer $n > 1$. Give an example of three non-identity $n \times n$ matrices A , B , and C so that the product ABC equals I_n , the $n \times n$ identity matrix; or explain why this is not possible.

Many answers are possible. For instance:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

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3. a) Find the 2×2 matrix for the following linear transformation using the standard basis for $\mathbb{R}^2$.

scale $\mathbb{R}^2$ by a factor of 5 and then rotate counterclockwise by 30°

second. first

$$\begin{pmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} 5\sqrt{3}/2 & -5/2 \\ 5/2 & 5\sqrt{3}/2 \end{pmatrix}$$

although, note that these linear transformations commute.

$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ for $\theta = 30^\circ$

b) Find the 3×3 matrix for the following linear transformation using the standard basis for $\mathbb{R}^3$.

rotate about the y -axis by $\pi/2$ counterclockwise

$$\begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Notice that if you strike out the y column and row, this yields:

$$\begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

a rotation by $\pi/2$ CCW in the x - z plane.

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4. a) Use row reduction to put the following real matrix in reduced row echelon form.

$$\begin{pmatrix} 2 & 4 & 0 & 6 \\ 2 & 4 & 1 & 10 \\ 3 & 6 & 1 & 13 \end{pmatrix} \xrightarrow{R_2 - R_1} \begin{pmatrix} 2 & 4 & 0 & 6 \\ 0 & 0 & 1 & 4 \\ 3 & 6 & 1 & 13 \end{pmatrix} \xrightarrow{\frac{1}{2}R_1} \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 3 & 6 & 1 & 13 \end{pmatrix}$$

$$\xrightarrow{R_3 - 3R_1} \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

↑ This is
in RREF.

b) Write the following linear system over $\mathbb{R}$ as an augmented matrix. Then find the general solution to the system by using row reduction and write the general solution in vector form.

$$3x + 2y - z = 4$$

$$x + 2y + z = 0$$

$$-x + 2y + 3z = -4$$

$$\xrightarrow{\quad} \begin{pmatrix} x & y & z & | & \\ 3 & 2 & -1 & | & 4 \\ 1 & 2 & 1 & | & 0 \\ -1 & 2 & 3 & | & -4 \end{pmatrix}$$

$$\xrightarrow{R_3 + R_2} \begin{pmatrix} 3 & 2 & -1 & | & 4 \\ 1 & 2 & 1 & | & 0 \\ 0 & 4 & 4 & | & -4 \end{pmatrix} \xrightarrow{R_1 - 3R_2} \begin{pmatrix} 0 & -4 & -4 & | & 4 \\ 1 & 2 & 1 & | & 0 \\ 0 & 4 & 4 & | & -4 \end{pmatrix}$$

$$\xrightarrow{R_1 + R_3} \begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 1 & 2 & 1 & | & 0 \\ 0 & 4 & 4 & | & -4 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_1} \begin{pmatrix} 1 & 2 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 4 & 4 & | & -4 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 1 & | & 0 \\ 0 & 4 & 4 & | & -4 \\ 0 & 0 & 0 & | & 0 \end{pmatrix} \xrightarrow{\frac{1}{4}R_2} \begin{pmatrix} 1 & 2 & 1 & | & 0 \\ 0 & 1 & 1 & | & -1 \\ 0 & 0 & 0 & | & 0 \end{pmatrix} \xrightarrow{R_1 - 2R_2} \begin{pmatrix} x & y & z & | & \\ 1 & 0 & -1 & | & 2 \\ 0 & 1 & 1 & | & -1 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$$\text{So } \begin{matrix} x = 2 + z \\ y = -1 - z \\ z = z \end{matrix} \quad \text{or} \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

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5. a) Compute the inverse of the following matrix.

$$A = \begin{pmatrix} 0 & 0 & 8 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 0 & 0 & 8 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 \end{array} \right)$$

$R_1 \leftrightarrow R_2$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 8 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 \end{array} \right)$$

$R_2 \leftrightarrow R_3$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 8 & 1 & 0 & 0 \end{array} \right)$$

$\frac{1}{8}R_2, \frac{1}{8}R_3$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & 1/8 & 0 & 0 \end{array} \right)$$

$$\text{so } A^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1/2 \\ 1/8 & 0 & 0 \end{pmatrix}$$

$$\text{Check: } \begin{pmatrix} 0 & 0 & 8 \\ 1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1/2 \\ 1/8 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

b) Let A and B be the following matrices. Compute the inverse matrix of the product AB ; or explain why this is not possible.

$$A = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Two approaches:

① Compute AB , find its inverse.

$$AB = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -2 & -2 \end{pmatrix}$$

$$\text{so } (AB)^{-1} = \frac{1}{-4} \begin{pmatrix} -2 & 0 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 1/2 & 0 \\ -1/2 & -1/2 \end{pmatrix}$$

or ② Find A^{-1} , B^{-1} . Then compute $(AB)^{-1} = B^{-1}A^{-1}$.

$$A^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & -1/2 \end{pmatrix}, \quad B^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \quad B^{-1}A^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1/2 & 0 \\ 0 & -1/2 \end{pmatrix} = \begin{pmatrix} 1/2 & 0 \\ -1/2 & -1/2 \end{pmatrix}$$

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6. a) Give a basis for the column space of A .

$$A = \begin{pmatrix} 2 & 5 & 3 & 1 & 4 \\ 4 & 12 & 6 & 4 & 12 \\ 6 & 10 & 9 & -2 & 2 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 2 & 5 & 3 & 1 & 4 \\ 0 & 2 & 0 & 2 & 4 \\ 6 & 10 & 9 & -2 & 2 \end{pmatrix}$$

$$\xrightarrow{R_3 - 3R_1} \begin{pmatrix} 2 & 5 & 3 & 1 & 4 \\ 0 & 2 & 0 & 2 & 4 \\ 0 & -5 & 0 & -5 & -10 \end{pmatrix} \xrightarrow{\substack{R_3 \div -5 \\ R_2 \div 2}} \begin{pmatrix} 2 & 5 & 3 & 1 & 4 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 & 2 \end{pmatrix}$$

$$\xrightarrow{R_3 - R_2} \begin{pmatrix} 2 & 5 & 3 & 1 & 4 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \text{ REF} \Rightarrow \text{Since the first two columns are a basis for the column space of the last matrix, the first two columns of the original matrix}$$

b) Give an example of a basis for $\mathbb{R}^4$ containing the vectors $(0, 0, 1, 2)^T$ and $(0, 0, 2, 3)^T$ or explain why this is not possible.

are a basis for its column space:
basis for $\text{col } A = \left\{ \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}, \begin{pmatrix} 5 \\ 12 \\ 10 \end{pmatrix} \right\}$

$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 2 \\ 3 \end{pmatrix}$ are linearly independent,

since they are not scalar multiples of each other. So they can belong to a basis for $\mathbb{R}^4$. Such a basis must contain 4 vectors. So for instance: (that are linearly independent and span $\mathbb{R}^4$)

$$\left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

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7. a) Using the standard basis, a linear transformation is given by the matrix A . Write the same linear transformation using v_1 and v_2 as an ordered basis for $\mathbb{R}^2$, where these vectors are given in the standard basis.

$$A = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}, \quad v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

A is the matrix for the linear transformation in the standard basis, S . So to write the linear transformation in the basis $B = v_1, v_2$, we first convert from working in B to working in S , then do the transformation working in S , then convert back to B . So the desired matrix $M = [I]_{SB} A [I]_{BS} = [I]_{SB}^{-1} A [I]_{BS}$

$$= \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \boxed{\begin{pmatrix} -1 & 0 \\ 2 & 2 \end{pmatrix}}$$

$$[I]_{SB} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \text{ because this is the matrix that takes } \begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } \begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

b) Give an example of three distinct real 3×3 matrices that are all similar to each other.

Check: eg. $\begin{pmatrix} -1 & 0 \\ 2 & 2 \end{pmatrix}_B \begin{pmatrix} 1 \\ 0 \end{pmatrix}_B = \begin{pmatrix} -1 \\ 2 \end{pmatrix}_B$ And $\begin{pmatrix} -1 \\ 2 \end{pmatrix}_B = \begin{pmatrix} 1 \\ 3 \end{pmatrix}_S$ ie. v_1 written in $B \mapsto v_1$ written in S
 $\begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}_S \begin{pmatrix} 1 \end{pmatrix}_S = \begin{pmatrix} 1 \\ 3 \end{pmatrix}_S$ since $-1 \begin{pmatrix} 1 \end{pmatrix} + 2 \begin{pmatrix} 1 \end{pmatrix} = \begin{pmatrix} 1 \end{pmatrix}$ + v_2 written in $B \mapsto v_2$ written in S .

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \text{ and } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

are all reflections in a plane, just different planes.

$$\text{or } \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\text{and } \begin{pmatrix} 1 & -8 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 8 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 8 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

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8. a) Compute the determinant of this matrix:

$$\begin{pmatrix} 0 & 1 & 0 \\ 2 & 1 & 3 \\ -1 & 2 & 2 \end{pmatrix}$$

② Cofactor expansion:

$$\begin{vmatrix} 0 & 1 & 0 \\ 2 & 1 & 3 \\ -1 & 2 & 2 \end{vmatrix} = -1 \begin{vmatrix} 2 & 3 \\ -1 & 2 \end{vmatrix} = -1(4+3) = -7$$

Two approaches:

① row reduction

$$\begin{pmatrix} 0 & 1 & 0 \\ 2 & 1 & 3 \\ -1 & 2 & 2 \end{pmatrix} \xrightarrow{R1 \leftrightarrow R2} \begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 0 \\ -1 & 2 & 2 \end{pmatrix} \xrightarrow{R3 + \frac{1}{2}R1} \begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 0 \\ 0 & 2.5 & 3.5 \end{pmatrix}$$

$$\xrightarrow{R3 - 2.5R2} \begin{pmatrix} 2 & 1 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 3.5 \end{pmatrix}$$

Since this matrix is upper triangular, its determinant is $(2)(1)(3.5) = 7$.

The row moves we applied to get here from the original matrix do not affect the determinant, except for the single row swap. So the original matrix has determinant -7 .

b) Give two distinct examples, A and B , of a 3×3 matrix with determinant 14. Then find $\det(A^{-1})$ and $\det(A^2 B^{-1})$.

Diagonal matrices have easy to compute determinants.

So for instance, $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 7 \end{pmatrix}$ $B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$$\det(A^{-1}) = \frac{1}{\det A} = \frac{1}{14}$$

(Note: A is invertible, since it has nonzero determinant. So A^{-1} makes sense.)

$$\det(A^2 B^{-1})$$

$$= \det(A) \cdot \det(A) \cdot \det(B^{-1})$$

$$= 14 \cdot 14 \cdot \frac{1}{14} = 14.$$